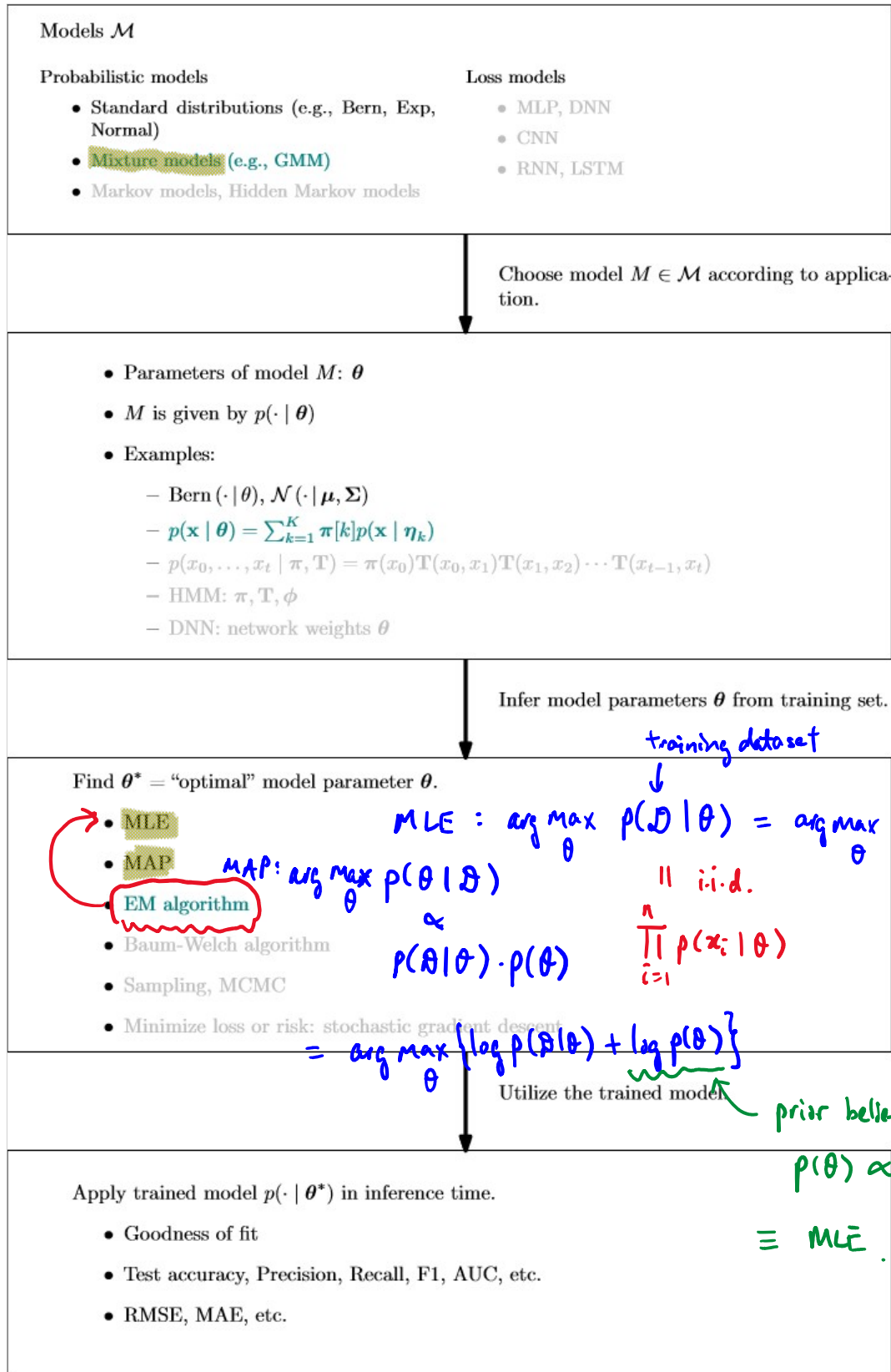




overview
week 3



training dataset

$$\text{MLE} : \arg \max_{\theta} p(\mathcal{D} | \theta) = \arg \max_{\theta} \log p(\mathcal{D} | \theta)$$

$$\text{MAP} : \arg \max_{\theta} p(\theta | \mathcal{D})$$

$$\propto p(\mathcal{D} | \theta) \cdot p(\theta) \quad \prod_{i=1}^n p(x_i | \theta)$$

$$= \arg \max_{\theta} \{ \log p(\mathcal{D} | \theta) + \log p(\theta) \}$$

$$\begin{aligned} & \log p(\mathcal{D} | \theta) \\ & \propto \sum_{i=1}^n \log p(x_i | \theta) \\ & \frac{\partial}{\partial \theta} : \sum_{i=1}^n \frac{\partial}{\partial \theta} \log p(x_i | \theta) \end{aligned}$$

prior belief

$$\begin{aligned} p(\theta) & \propto 1 \\ & \equiv \text{MLE} \\ \log p(\theta) & = \log \text{const} \\ \frac{\partial}{\partial \theta} \log p(\theta) & = 0 \end{aligned}$$

parameter weight

$$p(x|\eta) \quad \text{overall model weight} \quad p(\cdot|\eta_1) \quad p(\cdot|\eta_2) \quad \dots \quad p(\cdot|\eta_K)$$

mixture model

$$p(x|\theta) = \pi(1)p(x|\eta_1) + \pi(2)p(x|\eta_2) + \dots + \pi(K)p(x|\eta_K) \quad \checkmark$$

$$\pi = (\pi(1), \pi(2), \dots, \pi(K))$$

$$\theta = (\eta_1, \eta_2, \dots, \eta_K, \pi) \quad \hookrightarrow \text{categorical distribution}$$

$$\pi(k) \geq 0, \quad \sum_{k=1}^K \pi(k) = 1$$

unobserved

$$= \sum_{k=1}^K \pi(k) p(x|\eta_k)$$

Latent variable z

$$z \in \{1, 2, \dots, K\} \sim \text{Cat}(\pi)$$

$$\pi(k) = p(z=k)$$

Gaussian mixture model: GMM

mean covariance

$$p(\vec{x}|\theta) = \sum_{k=1}^K \pi(k) N(\vec{x} | \vec{\mu}_k, \Sigma_k)$$

$$\theta = (\pi, (\mu_k, \Sigma_k)_{k=1}^K)$$

$$\frac{\partial}{\partial \theta} \log p(\theta|\theta) = \sum_{i=1}^n \log p(x_i|\theta)$$

$$= \sum_{i=1}^n \left[\frac{\partial}{\partial \theta} \log \left(\sum_{k=1}^K \pi(k) p(x_i|\eta_k) \right) \right]$$

observation

$$\max \left\{ \underbrace{\pi(1) N(x | \mu_1, \sigma_1^2 I)}_{\substack{\downarrow \\ 1, 2, \dots, 0 \\ 1, 2}} + \underbrace{\pi(2) N(x | \mu_2, \sigma_2^2 I)}_{\text{finite } \geq 0} \right\}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2\sigma_1^2} (x - \mu_1)^2}$$

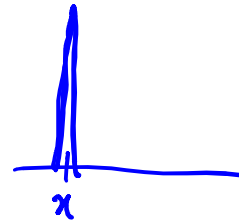
finite > 0 .

$$\boxed{\mu_1 = x}$$

$$= \frac{1}{\sqrt{\pi} \sigma_1}$$

$$\max_{\sigma_1} \left\{ \boxed{\frac{1}{\sqrt{2\pi} \sigma_1}} + \text{finite value } (\mu_2, \sigma_2) \right\}$$

$\sigma_1 \rightarrow 0$



$$\begin{matrix} \pi(1) & \mu_1 & \Sigma_1 \\ 0.3 & N(x | 0, 1) \end{matrix} + \begin{matrix} \pi(2) & \mu_2 & \Sigma_2 \\ 0.7 & N(x | 2, 5) \end{matrix} \quad \theta_1 = \left(\begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}, \begin{pmatrix} 0, 1 \\ 2, 5 \end{pmatrix} \right)$$

$$0.7 N(x | 2, 5) + 0.3 N(x | 0, 1) \quad \theta_2 = \left(\begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}, \begin{pmatrix} 2, 5 \\ 0, 1 \end{pmatrix} \right)$$

$$\log p(x_1, x_2, \dots, x_n | \theta)$$

$$\log p((x_1, z_1), (x_2, z_2), \dots, (x_n, z_n) | \theta)$$

$$\stackrel{i.i.d.}{=} \sum_{i=1}^n \log p(x_i, z_i | \theta)$$

$$= \sum_{i=1}^n \log(\pi(z_i) p(x_i | z_i, \theta))$$

$$= \sum_{i=1}^n \log(\pi(z_i) p(x_i | \eta_{z_i}))$$

$$= \sum_{i=1}^n \log \pi(z_i) + \sum_{i=1}^n \log p(x_i | \eta_{z_i})$$

$$= \sum_{i=1}^n \log \pi(z_i) + \sum_{i=1}^n \log p(x_i | \eta_{z_i})$$

$$\frac{\partial}{\partial \theta} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log \pi(z_i) + \sum_{i=1}^n \frac{\partial}{\partial \theta} \log p(x_i | \eta_{z_i}) = 0$$

$y_i = (x_i, z_i)$ complete data observed x_i
incomplete data

$x_i = T(y_i) = \text{1st coordinate of } y_i$

↗ "guesstimate" of θ

$$\mathbb{E}_{y \sim p(\cdot | x, \hat{\theta})} [\log p(y | \theta) | \vec{x}, \hat{\theta}]$$

$p(y | \theta) = \pi(z) p(x | \eta_z)$
↑
 $\theta = (\pi, (\eta_k)_{k=1}^K)$

Expectation-Maximization (EM) Algorithm

- $\theta^{(0)}$

- Iterate:

- E step:

guess
↓

$$Q(\theta | \theta^{(m)}) = \mathbb{E}_{y \sim p(y | x, \theta^{(m)})} [\log p(y | \theta) | x, \theta^{(m)}]$$

$$= \int_{-\infty}^{\infty} \log p(y | \theta) p(y | x, \theta^{(m)}) dy$$

- M step:

$$\theta^{(m+1)} = \arg \max_{\theta} Q(\theta | \theta^{(m)})$$

obs. latent
 ↓ ↓

$$\vec{y} = (y_i)_{i=1}^n = ((x_i, z_i))_{i=1}^n \quad \vec{x} = (x_i)_{i=1}^n$$

$$Q(\theta | \theta^{(m)}) = \mathbb{E} [\log p(\vec{y} | \theta) | \vec{x}, \theta^{(m)}]$$

$$= \mathbb{E} \left[\sum_{i=1}^n \log p(x_i, z_i | \theta) \mid \vec{x}, \theta^{(m)} \right]$$

$$\mathbb{E}[X]$$

$$= \sum_{i=1}^n \mathbb{E} [\log p(x_i, z_i | \theta) \mid \vec{x}, \theta^{(m)}]$$

$$= \sum_k p(z=k) \mathbb{E}[X | z=k]$$

$$= \sum_{k=1}^K \underbrace{p(z_i=k | \vec{x}, \theta^{(m)})}_{r_{ik}^{(m)}} \mathbb{E} [\log p(x_i, z_i | \theta) \mid \underline{z_i=k}, \vec{x}, \theta^{(m)}]$$

responsibility

$$r_{ik}^{(m)} = p(z_i=k | \vec{x}, \theta^{(m)}) = p(z_i=k | x_i, \theta^{(m)})$$

Bayes' rule.

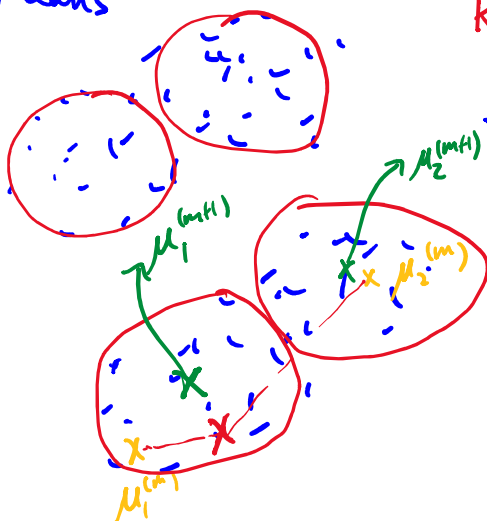
$$= \frac{p(z_i=k | \theta^{(m)}) p(x_i | z_i=k, \theta^{(m)})}{p(x_i | \theta^{(m)})}$$

$$= \frac{\pi^{(m)}(k) N(x_i | \mu_k^{(m)}, \Sigma_k^{(m)})}{\sum_{k'} \pi^{(m)}(k') N(x_i | \mu_{k'}^{(m)}, \Sigma_{k'}^{(m)})}$$

$$\begin{aligned} & \log p(x_i, z_i=k | \theta) \\ &= \log \pi(z_i=k) p(x_i | \mu_k, \Sigma_k) \\ &= \log \pi(k) + \log N(x_i | \mu_k, \Sigma_k) \end{aligned}$$

K-Means

K is known



$$\mu_k^{(mH)} \Sigma_k = \sigma^2 I$$

$$\pi(k) = \frac{1}{K}$$

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Bayes' rule

$$p(\theta | \mathcal{D}) = \frac{p(\mathcal{D} | \theta) p(\theta)}{p(\mathcal{D})}$$

$$\log p(\theta | \mathcal{D}) = \log p(\mathcal{D} | \theta) + \log p(\theta)$$